

TRUONG Le Gia Khanh

The University of Sydney, Australia

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Education

The University of Sydney

Sydney, Australia

Doctor of Philosophy in Mathematics and Statistics

3/2026 - present

- PhD topic: **Manifold learning in Generative models**
- Supervisors: **Prof. Georg Gottwald, Dr. Tiangang Cui**

Ho Chi Minh City University of Education

Ho Chi Minh City, Vietnam

Bachelor's Degree in Mathematics Education (Honours)

9/2021 - 5/2025

CGPA: 3.89/4.00

Valedictorian of both Program and Faculty

- Best Thesis Award (perfect score) for topic **Boundedness of Singular Stochastic Integral Operator**
- Top 5 of Faculty (4/4 years) and Top 1% of University (3/4 years)

Tran Dai Nghia High School for the Gifted

Ho Chi Minh City, Vietnam

High School Diploma: Specialist in Mathematics

9/2018 - 5/2021

CGPA: 9.3/10.0

- Top 2% national students in National High School Graduation Exam (27.05/30.0).
- Highest Score in Mathematics Subjects (3/3 years).

Research Experience

Quantum AI Lab, Phenikaa Institute for Advanced Study

Vietnam, Remote

Quantum Researcher (part-time)

5/2025 - 5/2026

- Project 1 (under Global Industry Challenge 2026): Predicting Secured Overnight Financing Rate (SOFR) Forward Dynamics via Quantum Reservoir Computing
- Project 2 (under Global Industry Challenge 2026): Leveraging the Generative Quantum Eigensolver (GQE) for the Scalable Design and Fabrication of Topological EUV Semiconductor Devices

IVS Joint Stock Company (Individual Systems)

Vietnam

Robotics Researcher (full time)

8/2025 - 02/2026

- Project 1: Shortest path and path planning for single mobile robot in two-dimensional plane.
- Project 2: Swarm formation control for multi robots system in two-dimensional plane.
- Supervisor: **Prof. Phan Thanh An, Dr. Dau The Phiet**

Institute of Applied Artificial Intelligence and Robotics (IAAIR)

USA, Remote

Quantum Internship Researcher (full time)

7/2025 - 04/2026

- Project 1: Benchmarking Quantum Machine Learning Algorithms from Mathematical Perspective
- Project 2: Learning Dataset Symmetries for Equivariant Quantum Convolutional Neural Networks on Biomedical Signals
- Supervisor: **Dr. Srikanth Thudumu, PhD Student Hung Du**

Vietnam Academy of Science and Technology

Ha Noi City, Vietnam

Potential Researcher (full time)

6/2024 - 8/2024

- Work under the grants of Vingroup Innovation Foundation and Simons Foundation.
- Project: Manifold Learning Methods and Applications.
- Supervisor: **Dr. Luu Hoang Duc**

Collaborative Mathematical Research for Undergraduate Students

Hybrid

Potential Researcher (full time)

9/2023-6/2024

- Work under the grant "Faculty Global Research with Undergraduates" from the University of Virginia Center for Global Inquiry and Innovation.
- Project: Random walk on Cayley graphs and unimodular random graphs.
- Supervisor: **Prof. Ben Hayes**

Ho Chi Minh City University of Education

Ho Chi Minh City, Vietnam

Undergraduate Researcher (full time)

11/2021-4/2022, 11/2023-4/2024, and

11/2024-4/2025

- Work under the grant of HCMUE Foundation for Science and Technology in 5 projects.
- Project: Singular stochastic integral operator. From September 2024 to April 2025. Advisor: **Msc. Nguyen Thanh Tung, Dr. Nguyen Ngoc Trong.**
- Project: Computation of extension groups between truncated polynomial functors in the category of strict polynomial functors. From September 2024 to April 2025. Advisor: **Dr. Nguyen Le Chi Quyet.**
- Project: Layer potential methods for boundary value problems. From September 2023 to April 2024. Advisor: **Dr. Nguyen Ngoc Trong.**
- Project: A well-balanced numerical scheme for the Shear Shallow Water model with bathymetry. From September 2023 to April 2024. Advisor: **Dr. Dao Huy Cuong.**
- Project: Mobile learning in mathematics education. From September 2021 to April 2022. Advisor: **Dr. Tang Minh Dung.**

Academic Activities

PUBLICATIONS

Parabolic Littlewood-Paley Inequality of Hilbert-valued Stochastic Integral with Stochastic Hörmander Condition
Truong Le Gia Khanh, Nguyen Ngoc Huu An
HCMUE Journal of Science, Accepted (2025) [[preprint pdf](#)]

Homology Computation of Bicomplexes between some Polynomial Functors
Truong Le Gia Khanh, Nguyen Le Chi Quyet
HCMUE Journal of Science, Accepted (2025) [[preprint pdf](#)]

A well-balanced numerical scheme based on a stationary contact jump for the shear shallow water model with bathymetry
Truong Le Gia Khanh, Dao Huy Cuong
Submitted (2024) [[preprint pdf](#)]

Weak Harnack Inequality for Schrödinger-type Operator
Nguyen Duc Trung, Truong Le Gia Khanh, Nguyen Ngoc Huu An, Nguyen Trong Nhan, Nguyen Thuy Tien
HCMUE Journal of Science 21.2 (2024) 256 [[link](#)]

Mobile learning in mathematics education: A systematic literature review of empirical research
Tang Minh Dung, Nguyen Thi Ngoc Cam, Truong Le Gia Khanh, Vo Khoi Nguyen, Nguyen Thai Hung, Le Tuan Khai, Bui Ngoc Huy, Tran Thanh Nha, Nguyen Thuy Tien
EURASIA Journal of Mathematics, Science and Technology Education 19.5 (2023) em2268. Modestum [[link](#)]

Achievements

ACADEMIC

- 2024 **First Prize**, *Competition for Designing Teaching Aids and Digital Learning Materials*, Ho Chi Minh City University of Education
- 2022 **Second Prize in Algebra**, *National Mathematical Olympiad (Table of Advanced Universities)*, Vietnam Mathematical Society
- 2021 **First Prize**, *City Excellent Students Contest in Solving Mathematics with Calculator (Table of High Schools)*, Department of Education and Training, Ho Chi Minh City
- 2021 **Second Prize**, *City Excellent Students Contest in Mathematics (Table of Advanced High Schools)*, Department of Education and Training, Ho Chi Minh City
- 2018 **Second Prize**, *City Excellent Students Contest in Mathematics (Table of Secondary Schools)*, Department of Education and Training, Ho Chi Minh City

SCHOLARSHIP

- 2026-2029 **Fully funded**, *Postgraduate Research Scholarship in Mathematics and Statistics*, The University of Sydney
- 2026-2029 **Fully funded**, *Faculty of Science Research Tuition Fee Scholarship*, The University of Sydney
- 2025 **Merit based**, *Never Give Up Scholarships for Talented Students*, Sponsorship program for talented student, 1800\$ AUD
- 2024 **Fully funded**, *Research Internship Scholarship*, Vietnam Academy of Science and Technology, 300\$ AUD
- 2021-2025 **Merit based**, *Outstanding Academic Scholarships*, Ho Chi Minh City University of Education, 800\$ AUD per year.
- 2022-2023 **Merit based**, *Paris - Benin - Vietnam Vallet Scholarships*, Fondation de France, 1600\$ AUD
- 2018-2021 **Merit based**, *Top 30% Highest Math GPA of Mathematics Honours class*, Tran Dai Nghia High school for the Gifted, 150\$ AUD per year

Skills

Standard Programming	Python (TensorFlow, PyTorch), Matlab
Quantum Programming	Qiskit, Pennylane, Cirq
Language	English, Vietnamese (Native)

Reference

Professor Georg Gottwald

- PhD project Supervisor
- School of Mathematics and Statistics
 - The University of Sydney, Australia
 - ✉ georg.gottwald@sydney.edu.au

Professor Ben Hayes

- Polymath REU Research Advisor (9 months)
- Department of Mathematics
 - University of Virginia, USA
 - ✉ brh5c@virginia.edu

Dr. Luu Hoang Duc

VAST Research Advisor (4 months)

- Max-Planck-Institut für Mathematik in den Naturwissenschaften, Leipzig, Germany & Institute of Mathematics, Vietnam Academy of Science and Technology, Vietnam
- ✉ lhduc@math.ac.vn, duc.luu@mis.mpg.de

Senior Lecturer Nguyen Ngoc Trong

Thesis and Research Advisor (12 months), Lecturer (3 courses) during undergraduate

- Department of Analysis
- Member of Group of Analysis and Applied Mathematics (GAAM) [\[link\]](#)
- Ho Chi Minh City University of Education, Vietnam Ministry of Education and Training
- ✉ trongnn@hcmue.edu.vn